Stochastic Processes

Finite Horizon MDP Problems

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Autumn, 2022

Outline

1. Review
2. Linear Quadratic Problem
3. Inventory Control
4. Optimal Stopping Problems
5. Scheduling Problems
6. Summary

MDP Problems

- State x_k of the system
- Action u_k to be selected at stage k
- State transition with random par. w_k

$$x_{k+1} = f_k(x_k, u_k, w_k), \qquad k = 0, 1, \ldots, N-1$$

  or described implicity by controlled transition prob. $p_{ij}(u)$
- Per-stage cost $g_k(x_k, u_k, w_k), k = 0, 1, \ldots, N-1$ and $g_N(x_N)$
- Objective: find optimal $\pi = \{\mu_0, \mu_1, \ldots, \mu_{N-1}\}$ for

$$\min_{\pi} E\left\{\sum_{k=0}^{N-1} g_k(x_k, \mu_k(x_k), w_k) + g_N(x_N)\right\}$$

DP Algorithm

- Start with
$$J_N(x_N) = g_N(x_N),$$
and go backwards for $k = N-1, N-2, \cdots, 0$
$$J_k(x_k) = \min_{u_k} E_{w_k} \left\{ g_k(x_k, u_k, w_k) + J_{k+1}\left(f_k(x_k, u_k, w_k)\right) \right\},$$
where $\mu_k^*(x_k)$ minimizes the right side for each x_k
- $J_0(x_0)$ is equal to the optimal cost $J^*(x_0)$, and the optimal policy is
$$\pi^* = \{\mu_0^*, \ldots, \mu_{N-1}^*\}$$

Structured Optimal Policy

- Deterministic DP Problems (w_k =Const)
  - Shortest path problems
  - Backward & forward algorithm
- Stochastic Problems (w_k is a random disturbance)
  - Perfect state information: exact value x_k is known in current stage k
- DP algorithm in general results in numerical solution
- For some well-structured problems, structured optimal policy exists
  - Closed-form
  - Threshold-based
  - $\cdots$

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Linear Quadratic Problems: Formulation

- Linear System

$$x_{k+1} = A_k x_k + B_k u_k + w_k$$

$$x_k \in \mathbb{R}^n, u_k \in \mathbb{R}^m$$

$w_k \in \mathbb{R}^n$ are independent random vectors where $E\{w_k\} = \mathbf{0}$.

- Quadratic cost

$$\underset{\substack{w_k \\ k=0,1,\ldots,N-1}}{E} \left\{ \sum_{k=0}^{N-1} (x_k^T Q_k x_k + u_k^T R_k u_k) + x_N^T Q_N x_N \right\}$$

- Q_k is positive semidefinite symmetric.
- R_k is positive definite symmetric.

Linear Quadratic Problems: Algorithm

- DP algorithm

$$J_N(x_N) = x_N^T Q_N x_N$$
$$J_k(x_k) = \min_{u_k} E\left\{x_k^T Q_k x_k + u_k^T R_k u_k + J_{k+1}(x_{k+1})\right\}$$

for $k = N-1, N-2, \cdots, 0$, where

$$x_{k+1} = A_k x_k + B_k u_k + w_k$$

Linear Quadratic Problems: Optimal Policy

- $J_k(x_k)$ is quadratic.
- $\mu_k^*(x_k) = L_k x_k$
- $L_k = -(B_k^T K_{k+1} B_k + R_k)^{-1} B_k^T K_{k+1} A_k$
- $J_k(x_k) = x_k^T K_k x_k + C_k$, K_k are positive semidefinite
- Dicrete-time Riccati equation

$$K_k = A_k^T (K_{k+1} - K_{k+1} B_k (B_k^T K_{k+1} B_k + R_k)^{-1} B_k^T K_{k+1}) A_k + Q_k$$

Riccati Equation and Asymptotic Behavior

- Stationary system with $A_k = A, B_k = B, Q_k = Q, R_k = R$
- The Riccati equation converges $\lim_{k\to-\infty} K_k = K$, where K is pos. definite, and is the unique (within the class of pos. semidefinite matrices) solution of the algebraic Riccati equation

$$K = A^T(K - KB(B^TKB + R)^{-1}B^TK)A + Q$$

- The corresponding steady-state controller $\mu^*(x) = Lx$, where

$$L = -(B^TKB + R)^{-1}B^TKA$$

is stable in the sense that the matrix $(A + BL)$ of the system

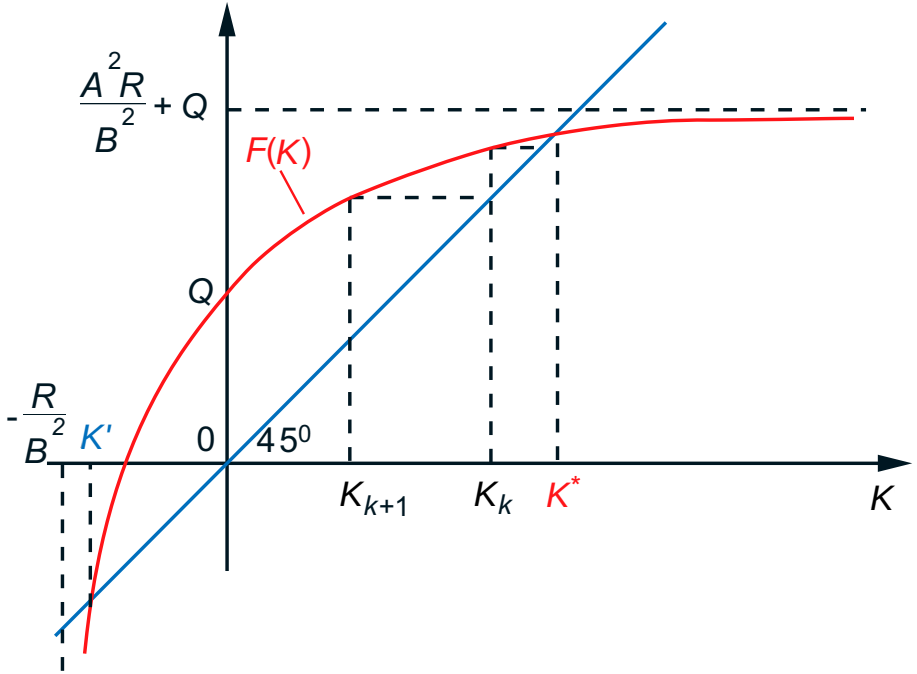
$$x_{k+1} = (A + BL)x_k + w_k$$

satisfies $\lim_{k\to\infty}(A + BL)^k = 0$.

Riccati Equation and Asymptotic Behavior

- In scalar systems, Riccati equation simplified to

$$K_k = \frac{A^2 R K_{k+1}}{B^2 K_{k+1} + R} + Q \triangleq F(K_{k+1})$$



- Two steady-state solutions of $K = F(K)$, only one is positive.

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Inventory Control: Model

- System $u_k \geq 0, x_k, w_k \in \mathbb{R}$

$$x_{k+1} = x_k + u_k - w_k, k = 0, 1, \ldots, N-1$$

- Cost

$$E\left\{\sum_{k=0}^{N-1}(cu_k + r(x_k + u_k - w_k))\right\}$$

- $r(x) = p\max(0, -x) + h\max(0, x), c < p$
- w_k is i.i.d.

Inventory Control: Derivation

- DP algorithm

$$J_N(x_N) = 0$$

$$J_k(x_k) = \min_{u_k \geq 0} \{cu_k + E\{r(x_k + u_k - w_k)\} + E\{J_{k+1}(x_k + u_k - w_k)\}\}$$

- Re-formulate as

$$J_k(x_k) \quad = \quad \min_{y_k \geq x_k} \{G_k(y_k)\} - cx_k$$

- where,

$$y_k = x_k + u_k$$

$$H(y) = E\{r(y - w_k)\}$$

$$G_k(y) = cy + H(y) + E\{J_{k+1}(y - w_k)\}$$

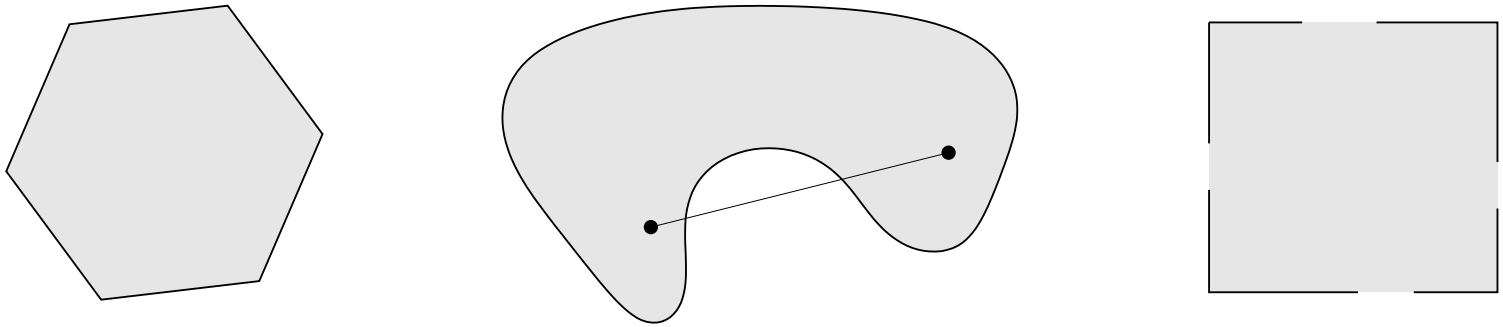
- Aim to prove G_k is convex and $\lim_{|x| \to \infty} G_k(x) \to \infty$

Appendix: Convexity

- Convex set $\mathcal{C}$

$$x_1, x_2 \in \mathcal{C}, 0 \leq \theta \leq 1, \Longrightarrow \theta x_1 + (1-\theta) x_2 \in \mathcal{C}$$

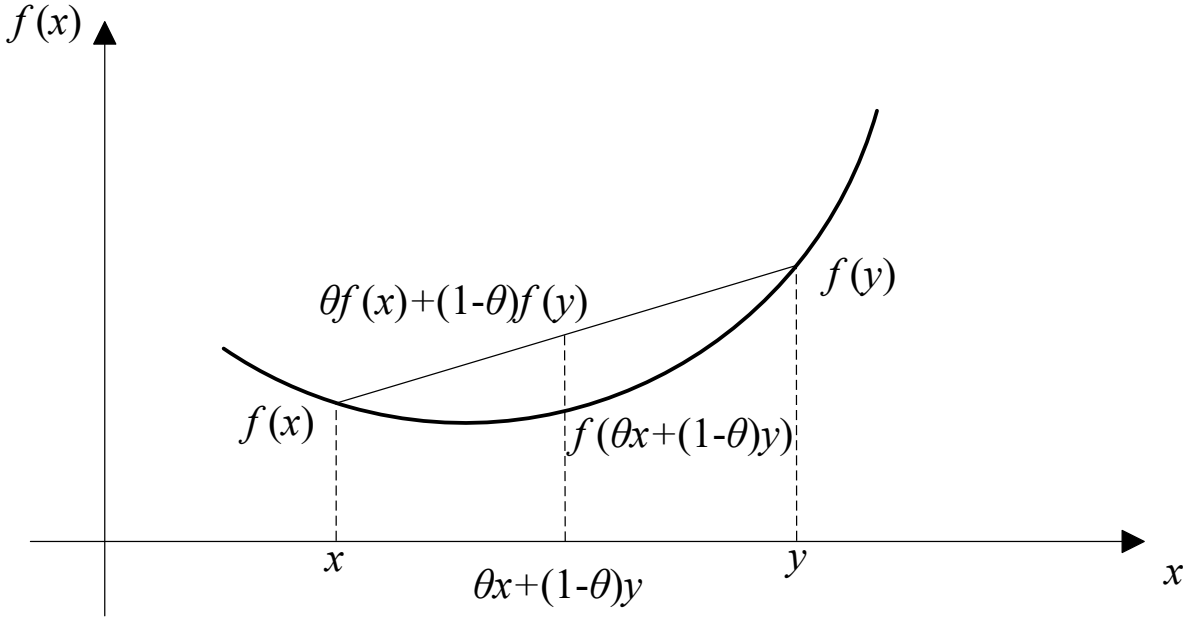
- Examples (one convex, two non-convex sets)



Appendix: Convexity

- Convex function $f : \mathcal{R}^n \to \mathcal{R}$
  - $\mathbf{dom} f$ is convex
  - for all $x, y \in \mathbf{dom} f, 0 \leq \theta \leq 1$

$$f(\theta x + (1-\theta)y) \leq \theta f(x) + (1-\theta) f(y)$$



Appendix: Convexity

- Examples

$$f(x) = ax + b, \quad f(x) = |x|, \quad f(x) = x^2$$

  - If f_1, f_2 are convex and $a \geq 0$, then

$$af_1(x), \quad f_1(x) + f_2(x)$$

    are convex (extends to infinite sums, integrals)
  - If $f_1, \cdots, f_m$ are convex, then

$$\max\{f_1(x), \cdots, f_m(x)\}$$

    is convex

- Convex function has a unique global minimum if it exists

Inventory Control: Optimal Policy

$$y_k = x_k + u_k$$

$$J_k(x_k) = \min_{y_k \geq x_k} \{G_k(y_k)\} - cx_k$$

- If G_k is convex and $\lim_{|x|\to\infty} G_k(x) \to \infty$, the optimal policy

$$\mu_k^*(x_k) = \begin{cases} S_k - x_k & \text{if} x_k < S_k \\ 0 & \text{if} x_k \geq S_k \end{cases}$$

where

$$S_k = \arg\min_{y\in\mathbb{R}} G_k(y)$$

Inventory Control: Proof

$$J_k(x) = \min_{y \geq x}\{G_k(y)\} - cx, \qquad J_N(x) = 0$$

$$G_k(y) = cy + H(y) + E\{J_{k+1}(y - w_k)\}$$

$$H(y) = E\{r(y - w_k)\}, \quad r(x) = p\max(0, -x) + h\max(0, x), c < p$$

- $cy + H(y)$ is convex and $\lim_{|y|\to\infty}(cy + H(y)) = \infty$
- J_k can be proved by induction
  - Prove the convexity of $G_{N-1}(x)$ and $\lim_{|x|\to\infty} G_{N-1}(x) = \infty$
  - Prove the convexity of $J_{N-1}(x)$ and $\lim_{|x|\to\infty} J_{N-1}(x) = \infty$
  - For $k = N-2, \cdots, 0$, assume the convexity of $J_{k+1}(x)$ and $\lim_{|x|\to\infty} J_{k+1}(x) = \infty$
    - Prove the convexity of $G_k(x)$ and $\lim_{|x|\to\infty} G_k(x) = \infty$
    - Then, prove the convexity of $J_k(x)$ and $\lim_{|x|\to\infty} J_k(x) = \infty$

Inventory Control: Proof

$$J_{N-1}(x_{N-1}) = \begin{cases} c(S_{N-1} - x_{N-1}) + H(S_{N-1}) & \text{if } x_{N-1} < S_{N-1} \\ H(x_{N-1}) & \text{if } x_{N-1} \geq S_{N-1} \end{cases}$$

$$J_k(x_k) = \begin{cases} c(S_k - x_k) + H(S_k) + E\{J_{k+1}(S_k - w_k)\} & \text{if } x_k < S_k \\ H(x_k) + E\{J_{k+1}(x_k - w_k)\} & \text{if } x_k \geq S_k \end{cases}$$

for all $k = N-2, \cdots, 0$

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Optimal Stopping Problems

- Problem Description
  - There exists a control that stops the evolution of the system
  - Decide at each stage whether to continue or stop the process
- Example: Asset Selling
  - A person has an asset (say a piece of land) to sell within N stages
  - In each stage k, a random amount of money w_k is offered
  - If the offer is accepted, invest the money at a fixed of rate of interest $r > 0$; otherwise, wait for the next offer
  - The last offer w_{N-1} must be accepted if every prior offer has been rejected

Asset Selling: Problem Formulation

- State x_k: the offer under consideration. $x_k = T$ means the asset has been sold.
- Action u_k: u^1 (sell) and u^2 (do not sell)
- State transition

$$\begin{aligned} x_{k+1} =& f_k(x_k, u_k, w_k) \\ =& \begin{cases} T & \text{if } x_k = T, \text{ or } x_k \neq T \text{ and } u_k = u^1(sell) \\ w_k & \text{otherwise} \end{cases} \end{aligned}$$

where $x_0 = 0$, w_k are i.i.d.

- Per-stage reward

$$g_N(x_N) = \begin{cases} x_N & \text{if } x_N \neq T, \\ 0 & \text{otherwise} \end{cases}$$

$$g_k(x_k, u_k, w_k) = \begin{cases} (1+r)^{N-k} x_k & \text{if } x_k \neq T \text{ and } u_k = u^1, \\ 0 & \text{otherwise} \end{cases}$$

Asset Selling: DP algorithm

- Start with

$$J_N(x_N) = g_N(x_N) = \begin{cases} x_N & \text{if } x_N \neq T, \\ 0 & \text{otherwise} \end{cases}$$

and go backwards using

$$\begin{aligned} J_k(x_k) =& \max_{u_k} E_{w_k} \left\{ g_k(x_k, u_k, w_k) + J_{k+1}\left(f_k(x_k, u_k, w_k)\right) \right\}, \\ =& \begin{cases} \max\left[(1+r)^{N-k} x_k, E\{J_{k+1}(w)\}\right] & \text{if } x_k \neq T, \\ 0 & \text{otherwise} \end{cases} \end{aligned}$$

Asset Selling: Optimal Policy

- Optimal policy

  accept the offer x_k if $x_k > \alpha_k$

  reject the offer x_k if $x_k < \alpha_k$

  where

$$\alpha_k = \frac{E\{J_{k+1}(w)\}}{(1+r)^{N-k}}$$

Asset Selling: Asymptotic Behavior

- Monotonicity property

$$\alpha_k \geq \alpha_{k+1}$$

  Let

$$V_k(x) = \frac{J_k(x)}{(1+r)^{N-k}}, \quad x \neq T$$

  Then

$$\alpha_k = \frac{E\{V_{k+1}(w)\}}{1+r}$$

  It is enough to show that $V_{k-1}(x) \geq V_k(x)$

- Proved by induction

Asset Selling: Proof

$$V_N(x) = J_N(x) = x$$

$$V_k(x) = \max\left[x, (1+r)^{-1}E\{V_{k+1}(w)\}\right], \quad k = 0, 1, \cdots, N-1$$

- For N and $N-1$, $V_{N-1}(x) \geq x = V_N(x)$
- Suppose $V_k \geq V_{k+1}$

$$\begin{aligned} V_{k-1}(x) &= \max\left[x, (1+r)^{-1}E\{V_k(w)\}\right] \\ &\geq \max\left[x, (1+r)^{-1}E\{V_{k+1}(w)\}\right] \\ &= V_k(x) \end{aligned}$$

Asset Selling: Asymptotic Behavior

- As $\alpha_k \geq 0$, the sequence $\{\alpha_k\}$ converges
- Notice that

$$V_k(x_k) = \max(x_k, \alpha_k)$$

- We have $(P(x) = \Pr(w < x))$

$$\alpha_k = \frac{E\{V_{k+1}(w)\}}{1+r} = \frac{P(\alpha_{k+1})}{1+r}\alpha_{k+1} + \frac{1}{1+r}\int_{\alpha_{k+1}}^{\infty} w \mathrm{d}P(w)$$

- Therefore, $\alpha_k \to \bar{\alpha}$ where

$$(1+r)\bar{\alpha} = P(\bar{\alpha})\bar{\alpha} + \int_{\bar{\alpha}}^{\infty} w \mathrm{d}P(w)$$

Asset Selling: Asymptotic Behavior

- Optimal stationary policy as $N \to \infty$

  accept the offer x_k if $x_k > \bar{\alpha}$

  reject the offer x_k if $x_k < \bar{\alpha}$

General Stopping Problems

- Consider a stationary system
- At time k, we may stop at cost $t(x_k)$ or choose a control $u_k \in U(x_k)$ and continue

$$J_N(x_N) = t(x_N)$$

$$J_k(x_k) = \min\left[t(x_k), \min_{u_k \in U(x_k)} E\left\{g(x_k, u_k, w_k) + J_{k+1}\left(x_{k+1}\right)\right\}\right]$$

- Optimal to stop at time k for state x in the set

$$T_k = \left\{x \middle| t(x) \leq \min_{u \in U(x)} E\left\{g(x, u, w) + J_{k+1}\left(f(x, u, w)\right)\right\}\right\}$$

which satisfies $T_0 \subset \cdots \subset T_k \subset T_{k+1} \subset \cdots \subset T_{N-1}$

- Requirement of stopping get loose

General Stopping Problems

Firstly, prove $J_k(x) \leq J_{k+1}(x)$

$$J_N(x_N) = t(x_N)$$

$$J_k(x_k) = \min\left[t(x_k), \min_{u_k \in U(x_k)} E\left\{g(x_k, u_k, w_k) + J_{k+1}\left(x_{k+1}\right)\right\}\right]$$

For $k = N - 1$,

$$\begin{aligned} J_{N-1}(x) =& \min\left[t(x), \min_{u \in U(x)} E\left\{g(x, u, w) + J_N\left(f(x, u, w)\right\}\right]\right. \\ \leq& t(x) = J_N(x) \end{aligned}$$

Suppose $J_k(x) \leq J_{k+1}(x)$,

$$\begin{aligned} J_{k-1}(x) =& \min\left[t(x), \min_{u \in U(x)} E\left\{g(x, u, w) + J_k\left(f(x, u, w)\right\}\right]\right. \\ \leq& \min\left[t(x), \min_{u \in U(x)} E\left\{g(x, u, w) + J_{k+1}\left(f(x, u, w)\right\}\right]\right. = J_k(x) \end{aligned}$$

General Stopping Problems

Proof of $T_0 \subset \cdots \subset T_k \subset T_{k+1} \subset \cdots \subset T_{N-1}$

$$T_k = \left\{ x \middle| t(x) \leq \min_{u \in U(x)} E\left\{ g(x,u,w) + J_{k+1}\left(f(x,u,w)\right)\right\} \right\}$$

For $\forall x \in T_{k-1}$,

$$\begin{aligned} t(x) &\leq \min_{u \in U(x)} E\left\{ g(x,u,w) + J_k\left(f(x,u,w)\right)\right\} \\ &\leq \min_{u \in U(x)} E\left\{ g(x,u,w) + J_{k+1}\left(f(x,u,w)\right)\right\} \end{aligned}$$

$\Longrightarrow x \in T_k \Longrightarrow T_{k-1} \subset T_k$

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Scheduling Problems

- We have a set of tasks to perform, the ordering is subject to optimal choice
- Costs depend on the order
- There may be stochastic uncertainty, and precedence and resource availability constraints
- Some problems can be solved by an interchange argument (start with some schedule, interchange two adjacent tasks,and see what happens)

Example: The Quiz Problem

- A list of N questions
- If question i is answered correctly (with prob. p_i), we receive a reward R_i; if not, the quiz terminates.
- Objective: choose order of questions to maximize expected reward

The Quiz Problem: MDP Formulation

- State: an order of correctly answered questions $x_k = (i_0, i_1, \cdots, i_{k-1})$
- Action: the question to be answered $u_k = i_k$
- State transition:

$$x_{k+1} = \begin{cases} (x_k, u_k), & \text{correct} \\ T, & \text{incorrect or } x_k = T \end{cases}$$

- Per-stage reward:

$$g_k(x_k, u_k, w_k) = \begin{cases} R_{i_k}, & \text{if } x_k \neq T \text{ and } i_k \text{ is correct,} \\ 0, & \text{otherwise} \end{cases}$$

The Quiz Problem: Interchange Argument

- For a given order $L = (i_0, i_1, \ldots, i_{N-1})$, the expected rewards is

$$p_{i_0}(1-p_{i_1})R_{i_0} + p_{i_0}p_{i_1}(1-p_{i_2})(R_{i_0}+R_{i_1}) + \ldots + \prod_{k=0}^{N-1} p_{i_k} \sum_{k=0}^{N-1} R_{i_k}$$

$$= p_{i_0}R_{i_0} + p_{i_0}p_{i_1}R_{i_1} + \ldots + \prod_{k=0}^{N-1} p_{i_k} R_{i_{N-1}}$$

- Consider two orders $L = (i_0, \ldots, i_{k-1}, i, j, i_{k+1}, \ldots, i_{N-1})$ and $L' = (i_0, \ldots, i_{k-1}, j, i, i_{k+1}, \ldots, i_{N-1})$

$$E\{\text{reward of } L\} = C_1 + p_{i_0} \cdots p_{i_{k-1}}(p_i R_i + p_i p_j R_j) + C_2$$

$$E\{\text{reward of } L'\} = C_1 + p_{i_0} \cdots p_{i_{k-1}}(p_j R_j + p_i p_j R_i) + C_2$$

The Quiz Problem: Interchange Argument

- If L is optimal, $E\{\text{reward of } L\} \geq E\{\text{reward of } L'\}$

$$\begin{aligned} &\Longrightarrow \quad p_i R_i + p_i p_j R_j \geq p_j R_j + p_i p_j R_i \\ &\Longrightarrow \quad \frac{p_i R_i}{1 - p_i} \geq \frac{p_j R_j}{1 - p_j} \end{aligned}$$

- Questions should be answered in decreasing order of $p_i R_i / (1 - p_i)$

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Summary

- MDP problems
- Linear-Quadratic
  - Linear policy
- Inventory Control
  - Threshold based policy
- Optimal Stopping Problems
  - Example: Asset selling
  - General stopping problems and the properties
- Scheduling Problems
  - Example: Quiz problem

Homework

Suppose that a person wants to sell a house and an offer comes at the beginning of each day. We assume that successive offers are independent and an offer is w_j with probability p_j, $j = 1, \ldots, n$, where w_j are given nonnegative scalars. Any offer not immediately accepted is not lost but may be accepted at any later date. Also, a maintenance cost c is incurred for each day that the house remains unsold. The objective is to maximize the price at which the house is sold minus the maintenance costs. Consider the problem when there is a deadline to sell the house within N days and characterize the optimal policy.